

Gabriel Isaac Alonso Serrato

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EDUCATION

Bachelor of Science in Computer Science, Concentration in Data Analytics
The University of Texas at El Paso (UTEP)

August 2021 - December 2025
GPA: 3.93 / 4.00

EXPERIENCE

Undergraduate Research Assistant

October 2024 – December 2025

NSF Funded CIELO-G Lab at The University of Texas at El Paso (UTEP)

- Initiated and led the development of the CIELO-G Learning Framework, a first-of-its-kind modular gamification platform for Earth-science education developed under the mentorship of Dr. Aaron Velasco, Dr. Katalina Salas, and PhD Marc Garcia. The framework addresses the absence of domain-specific gamification tools in geoscience.
- Designed and implemented four core subsystems: (1) an Explanation System for narrative-driven instructional slides with contextual dialogues; (2) a Clickable System that generalizes any object into an interactive multiple-choice element; (3) a Draggable System enabling drag-and-drop association and sorting tasks; and (4) a Mix-and-Match System supporting arbitrary pairwise mappings for concept-association exercises.
- Engineered the complete software architecture in Godot 4 (GDScript), including a centralized ModuleManager for flow control, scoring, and retry handling; a TransitionManager for cross-scene state management; and a reusable Manubar component for user feedback, progress tracking, and navigation across modules.
- Implemented a certificate-generation module that rewards full completion and reinforces learning retention through performance-based feedback.

Summer Geometry Initiative Research Fellowship

August 2 - 15, 2025

Massachusetts Institute of Technology (MIT) and University of Nevada, Reno

- Performed research under Dr. Ankita Shukla (University of Nevada, Reno) on the topological and geometric structures of AI-generated images, applying algebraic topology (persistent homology) and differential geometry to study latent-space organization in generative models (GANs, diffusion models).
- Implemented LID (Local Intrinsic Dimensionality) analysis with ResNet-18 feature embeddings to classify AI-generated (“fake”) vs real CIFAR-10 samples, producing class-wise distributions and significance tests.
- Estimated the minimal discriminative subspace dimension via PCA on deep features and developed an LID-based threshold classifier, jointly tuning neighbor count k and cutoff T^* ; achieved up to 0.65 accuracy (best $k \approx 318$, $T^* \approx 20.1$) with baseline ML classifiers (SVM, Logistic Regression) reaching $F1 \approx 0.60$.
- Built a visualization suite overlaying real/fake LID histograms with optimal thresholds, threshold-vs-accuracy curves across k , per-class LID bar plots with standard deviations and p -values, and confusion matrices for sensitivity analysis and reproducibility.

Summer Geometry Initiative Research Fellowship

July 19 - August 1, 2025

Massachusetts Institute of Technology (MIT) and Adobe Research

- Researched with Dr. Yi Zhou (Adobe Research) on Differentiable Mesh Reconstruction with DMesh++, exploring continuous probabilistic mesh representations for high-resolution reconstruction.
- Replaced slow virtual machine workflows with an optimized Windows Subsystem for Linux (WSL) setup, configuring CUDA and PyTorch for direct GPU access and achieving x2 faster DMesh++ training and visualization performance.
- Implemented a silhouette-aware tessellation function that improved mesh connectivity and continuity through multi-angle silhouette consistency, reducing reconstruction speed by an estimated 30% under identical 15-minute training budgets and settings.
- Proposed a 1:1 pixel-to-voxel optimization strategy, reformulating the mesh optimization process into voxel-position refinement for pixel-level reconstruction fidelity and gradient stability.
- Developed advanced visualization and comparative analysis tools integrating the Minimum-Ball and Weighted Delaunay Triangulation (WDT) algorithms, incorporating sigmoid-based probability modulation to dynamically assess connectivity

accuracy, tessellation continuity, and optimization stability as outlined in Section 3.1 -3.2 of the *DMesh++: An Efficient Differentiable Mesh for Complex Shapes* paper.

Summer Geometry Initiative Research Fellowship

July 12 - 18, 2025

Massachusetts Institute of Technology (MIT) and University of Toronto

- Conducted research on Adaptive Fluid-Particle Splitting with Dr. Eitan Grinspun and Ph.D. Jonathan Panuelo (University of Toronto), focusing on resolving under-sampled regions in Voronoi-based compressible fluid simulations at hypersonic speeds.
- Investigated Lagrangian Voronoi discretizations for compressible Euler flow, adapting the leak proof topology-preserving coupling framework to mitigate wake under-resolution and boundary separation.
- Implemented 4 Python prototypes using Taichi (GPU-accelerated kernels) and Houdini VEX scripts for particle-based Voronoi refinement, validating mass, momentum, and energy conservation across 2000+ frame sequences.
- Evaluated adaptive refinement methods to maintain sufficient and necessary leakproofness of the fluid domain, referencing Algorithm 1 and Section 4.2 (Clipped Voronoi Stitching) of the paper *Topology-Preserving Coupling of Compressible Fluids and Thin Deformables*.

Undergraduate Research Assistant

June 2023 - August 2023

Center of Nanoscience and Nanotechnology (CNyN) at the National Autonomous University of Mexico (UNAM)

- Conducted research under the supervision of Dr. Jonathan Guerrero on the molecular geometry of molybdenum disulfide (MoS_2), exploring over 2,000 candidate atomic configurations through computational modeling and polygon-based lattice generation, with the top three models submitted for supercomputer analysis.
- Implemented Python algorithms to dynamically generate triangular and hexagonal atomic lattices, computing centroids, interatomic distances, and nearest-neighbor alignments between molybdenum (Mo) and sulfur (S) atoms.
- Utilized Matplotlib and NumPy to visualize spatial distributions across thousands of simulated lattices, identifying patterns in symmetry, defect alignment, and molecular layer formation representative of MoS_2 structures.
- Developed a pipeline for analyzing, storing and exporting polygon-based molecular datasets for extended study of bonding geometry, stability, and material properties.

Google Tech Exchange Fellowship

January 2023 – May 2023

Google and The University of Texas at El Paso (UTEP)

- Selected as one of 150 students nationwide for the Google Tech Exchange Program, a semester-long academic initiative between Google and Hispanic Serving Institutes, focused on advancing representation in computer science and software engineering.
- Completed 7 credit hours of advanced coursework in Software Engineering and Applied Data Structures, designing scalable distributed systems using industry standards and best practices.
- Engineered a full-stack web application with Flask, hosted on Google Cloud Run and Cloud Storage Buckets. Designed and tested over 600 lines of code through unit tests and CI/CD pipelines, leveraging GitLab for version control and deployments.

PUBLICATIONS - CONFERENCE ABSTRACTS

- Alonso Serrato, G., Salas, K., Garcia, M., & Velasco, A. A. (2025, December 16). *Gaming to Better the Earth: A Community-Centered Framework for Earth Science Education Through Game Design*. Poster presented at the AGU Fall Meeting 2025, Session ED23B – *Advances and Progress in Fostering a Supportive Scientific Community for All II*, New Orleans, LA, USA. American Geophysical Union.

CONFERENCE PRESENTATIONS

American Geophysical Union (AGU) Conference – New Orleans, LA

December 2025

Research Poster Presentation — “*Gaming to Better the Earth: A Community-Centered Framework for Earth Science Education*”

High Impact and Innovative Teaching (HIIT) Conference – El Paso, TX

November 2025

Talk Presentation — “*A Gamification Framework for Democratizing Earth Science Education*”

SACNAS National Diversity in STEM Conference – Columbus, OH

October 2025

Research Poster Presentation — “*Gaming to Better the Earth: A Community-Centered Framework for Earth Science Education*”

PROJECTS & DELIVERABLES

CIELO-G Learning App [App Link]	September 2025
Developed and launched the official NSF-funded CIELO-G Web-app, which hosts and showcases playable, classroom-ready Earth-science modules built with my CIELO-G Learning Framework, a modular gamification platform.	
From Triangles to Gradients: Exploring DMesh++ [Article Link]	September 2025
Authored a research article during the MIT Summer Geometry Initiative (SGI) on the DMesh++ differentiable meshing framework, under the mentorship of Dr. Yi Zhou.	
Behavioral Deplasticization Animation [Video Link]	July 2025
Produced a 3D animation illustrating behavioral deplasticization concepts using Spline 3D and Adobe Premiere. Focused on clear visual storytelling to promote environmental awareness and behavior change.	
LEGO UTEP Animation [Video Link]	March 2025
Created a LEGO-style animation in Blender for the UTEP Geology Department's TCM Day, featuring a playful dinosaur theme.	
3D Educational Game Concept for Kids [Demo Link]	November 2024
Developed a 3D educational game with modular mini-games to enhance interactive learning, featuring Firebase for authentication, tracking, and analytics.	
Godot 4 Spaceship Controller [Demo Link]	October 2024
Built a 3D spaceship controller in Godot 4 with physics-based motion, thrust, and camera control. Optimized for gamepad gameplay.	
Infinite Runner (GMTK Game Jam) [Demo Link]	August 2024
Made an infinite runner for the 2024 GMTK Game Jam, with a character that changes size with speed and procedural obstacles.	
Unreal Engine 5 Voxel Character Demo [Demo Link]	February 2024
Developed a third-person voxel character in Unreal Engine 5 with interaction and puzzle systems.	

HACKATHONS & COMPETITIONS

Rube Goldberg Unreal Engine Challenge	May 2025
3rd Place (Senior Division) — <i>Built a physics-driven 3D Rube Goldberg Machine</i>	
STTE Foundation AI Hackathon	April 2025
1st Place — <i>Developed an AI-driven health platform with a trained LLM for accurate personalized guidance</i>	
Google DevFest Hackathon	September 2024
1st Place — <i>Developed a 3D Interactive Education Platform with Learning Performance Tracking</i>	
ACM-W AI Hackathon	March 2024
2nd Place — <i>Applied Reinforcement Learning for Optimizing Walkable Urban Map Navigation</i>	
Google Student Developer Club Hackathon	April 2023
1st Place — <i>Built a full-stack collaborative wiki for sharing green and sustainable solutions</i>	

AWARDS & HONORS

“What a Researcher Looks Like!” Feature Series	2025
The University of Texas at El Paso — <i>Recognized for contributions to creative and computational research initiatives</i>	
Houston Endowment Scholarship	2023 - 2025
The University of Texas at El Paso — <i>Merit-Based Award (\$2,000 USD per year up to 4 years)</i>	
College of Engineering Honors Program	2022 – 2025
The University of Texas at El Paso — <i>Recognized for Academic Excellence, Research Engagement, and Leadership Development</i>	
College of Engineering Dean’s List	2021 - 2025
The University of Texas at El Paso — <i>Academic Excellence Recognition</i>	
ORISE Scholarship	2022
Oak Ridge Institute for Science and Education — <i>“Communicating Quantum” for Science Education (\$1,000 USD)</i>	

LEADERSHIP & AFFILIATIONS

Society for Advancement of Chicanos and Native Americans in Science (SACNAS) The University of Texas at El Paso — <i>Active Member</i> Promoting diversity and inclusion in STEM through research and community engagement	2025
Art & Computer Science (Student Org.) The University of Texas at El Paso — <i>President</i> Lead a 15-member organization hosting weekly workshops on game development, geometry processing, computer graphics, and creative coding, bridging art and technology disciplines.	2024 – 2025
Apple Pathways Alliance Apple Inc. — <i>Selected Member</i> Participating in mentorship, technical training, and professional development focused on inclusion in technology.	2023 – 2025
Association for Computing Machinery (ACM) The University of Texas at El Paso — <i>Member</i> Engaged in computing projects, software development, and academic networking events.	2022 – 2023
Society of Hispanic Professional Engineers (SHPE) The University of Texas at El Paso — <i>Outreach Chair</i> Supported outreach initiatives for the local SHPE chapter, organizing STEM engagement and professional development programs.	2021 – 2022

RELEVANT COURSEWORK

SGI Introduction to Geometry Processing | Parallel Computing | Machine Learning | Data Mining | Computer Vision | Applied Data Structures | Advanced Object-Oriented Programming | Software Engineering I & II | Operating Systems | Database Systems | Computer Architecture

SKILLS

Programming Languages – Python, Java (Proficient); C++, C, C#, JavaScript, Bash (Intermediate)
Scientific Computing & Simulation – Numerical Methods, Finite-volume and Finite-difference Solvers, Geometry Processing, Differentiable Rendering, Parallel Computing, High-performance Computing
Machine Learning & Data Analysis – PyTorch, Scikit-learn, NumPy, Pandas, SciPy; Data Preprocessing, Model Training, Regression/Classification, Feature Extraction, Visualization, and Evaluation Metrics
Visualization & Rendering – OpenGL, Polyscope, Matplotlib, Blender; Scientific and Physically-based Visualization, Shader Programming, and 3D Data Rendering
Tools & Environments – Linux/WSL, Anaconda, Git, LaTeX, Google Cloud Run / Buckets, CI/CD Workflows
Speaking Languages – English (C2 - Professional Proficiency), Spanish — (Native)